

Quantum Phase Transitions: 0/0 at Absolute Zero

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The Removable Singularity Framework

Abstract

We show that quantum phase transitions are 0/0 removable singularities occurring at absolute zero ($T=0$), driven entirely by quantum fluctuations. The transverse field Ising model provides the canonical example: at the quantum critical point g_c , the order parameter $\langle \sigma_z \rangle = 0/0$, the energy gap closes ($\Delta=0$), and entanglement entropy diverges logarithmically. Remarkably, the 1D quantum Ising model has the SAME critical exponents ($\beta=1/8$, $\nu=1$) as the 2D classical Ising model -- a deep quantum-classical correspondence. The dynamical exponent $z=1$ (quantum) is the key difference from $z=0$ (classical). Topological entanglement entropy ($\gamma=\log 2$) and Berry phase (π) provide additional 0/0 signatures.

1. Transverse Field Ising Model

$$H = -J \cdot \sum (\sigma_z^i \cdot \sigma_z^{i+1}) - g \cdot \sum (\sigma_x^i)$$

The transverse field Ising model is the simplest quantum many-body system exhibiting a quantum phase transition. At $g=0$: fully ordered (all spins aligned in z). At $g=\infty$: fully disordered (all spins aligned in x). At $g_c = J = 1$: quantum critical point (0/0 removable singularity).

The order parameter: $\langle \sigma_z \rangle = (1 - g^2/J^2)^{1/8}$ for $g < J$, $\langle \sigma_z \rangle = 0$ for $g \geq J$. At $g = g_c$: $\langle \sigma_z \rangle = 0/0$ with critical exponent $\beta = 1/8$. This is the SAME exponent as the 2D classical Ising model -- a profound quantum-classical correspondence.

2. Energy Gap Closing

$$\Delta(g) = 2J \cdot |1 - g/J|$$

The energy gap between ground state and first excited state closes linearly at g_c . At $g = g_c$: $\Delta = 0$ (0/0 removable singularity). The gap closing is the quantum signature of the phase transition. In classical systems, the gap closes at T_c (thermal). In quantum systems, it closes at g_c (zero temperature, pure quantum fluctuations).

3. Entanglement Entropy Divergence

$$S \sim c/3 \cdot \log(L) \text{ at criticality, } c = 1/2$$

The entanglement entropy of a block of L sites diverges logarithmically at the quantum critical point. Away from criticality, it obeys an area law ($S \sim \text{constant}$). The central charge $c = 1/2$ identifies the Ising universality class. The logarithmic divergence is the quantum signature of long-range correlations at criticality.

4. Quantum-Classical Correspondence

The most remarkable result: the 1D quantum Ising model at $T=0$ has the SAME critical exponents as the 2D classical Ising model at $T=T_c$:

- $\beta = 1/8$ (both)
- $\nu = 1$ (both)
- $z = 1$ (quantum, Lorentz invariant)

- $z = 0$ (classical, no dynamics)

This correspondence arises because the transfer matrix of the 2D classical model is the Boltzmann weight $e^{-H_{dt}}$, which is the imaginary-time evolution operator of the 1D quantum model. The extra dimension (imaginary time) maps quantum to classical.

5. Quantum Critical Fan

At $T=0$: quantum fluctuations dominate. At $T>0$: thermal fluctuations dominate. The quantum critical fan is the region $T < |g - g_c|^{\nu z}$ where quantum critical behavior persists at finite temperature. Inside the fan: universal scaling, power-law correlations. Outside: exponential decay, conventional behavior.

6. Topological Signatures

The quantum phase transition has topological signatures:

- Topological entanglement entropy: $\gamma = \log(2)$ in ordered phase, 0 in disordered
- Berry phase: π in ordered phase, 0 in disordered phase
- At g_c : both jump discontinuously (0/0)

The $\mathbb{Z}_2$ topological order in the ordered phase is characterized by $\gamma = \log(2)$. This is the simplest example of topological order in a quantum phase transition.

7. Connections to Prior 0/0 Singularities

QPT	Prior Chapter	Connection
Ising $\beta=1/8$	Ch.36 Ising	Same exponents
Entanglement S	Ch.33 Entanglement	S diverges at g_c
Quantum consciousness	Ch.34 Consciousness	Penrose-Hameroff
AdS/CFT	Ch.32 Black holes	Quantum criticality
Quantum turbulence	Ch.37 Turbulence	Superfluid He
Quantum finance	Ch.38 Finance	Quantum models

8. Conclusion

Quantum phase transitions are the most fundamental 0/0 removable singularities. They occur at absolute zero, driven by pure quantum fluctuations with no thermal noise. The transverse field Ising model reveals a deep quantum-classical correspondence: 1D quantum has the same critical exponents as 2D classical ($\beta=1/8$, $\nu=1$), but with $z=1$ (Lorentz invariant) instead of $z=0$. Entanglement entropy, topological order, and Berry phase provide additional 0/0 signatures. The quantum 0/0 is the most fundamental because it occurs at $T=0$, where quantum fluctuations are the ONLY source of criticality.

References

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